Дообучение LLM

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План

1. Работа с промптами
2. PEFT
3. Практика

Работа с промптами

Работа с промптами: Chain-of-Thought reasoning

Standard Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ✗

Chain-of-Thought Prompting

Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. 5 + 6 = 11. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had 23 - 20 = 3. They bought 6 more apples, so they have 3 + 6 = 9. The answer is 9. ✓

When asked to “think”, the model gives the right answer

Reasoning-модели

Модели можно дообучать для ризонинга:

- COT-reasoning: модель дообучают на датасетах, в которых показаны рассуждения
- RLHF: люди оценивают reasoning модели и дообучение идет при помощи RL-модели, поощряющей хорошие рассуждения
- Также для ризонинг-моделей задаются системные промпты, чтобы побуждать модель рассуждать

Работа с промптами: self-consistency

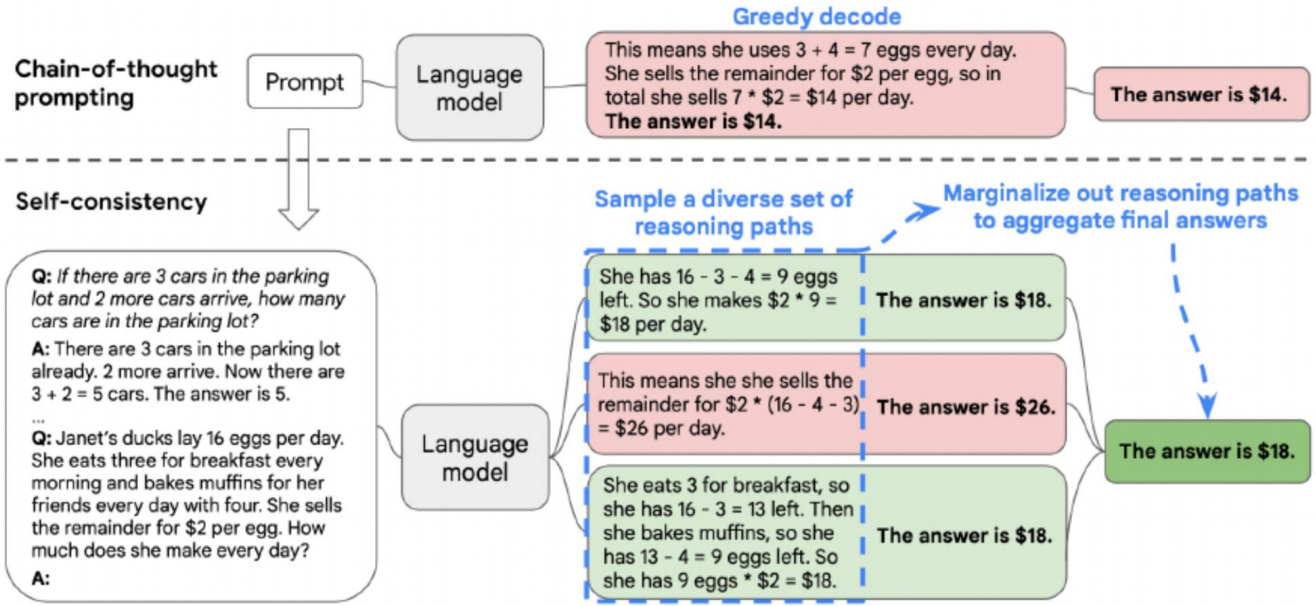
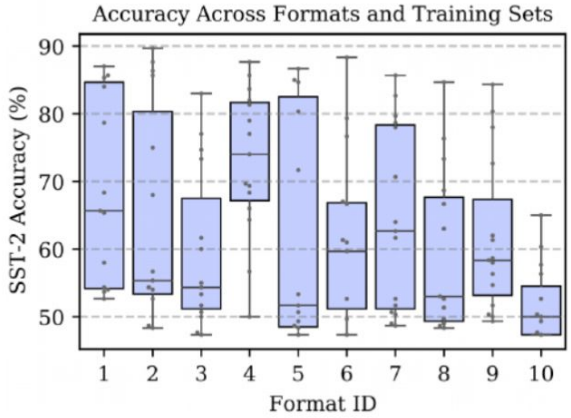


Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the "greedy decode" in CoT prompting by sampling from the language model's decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Работа с промптами: влияние формата промпта

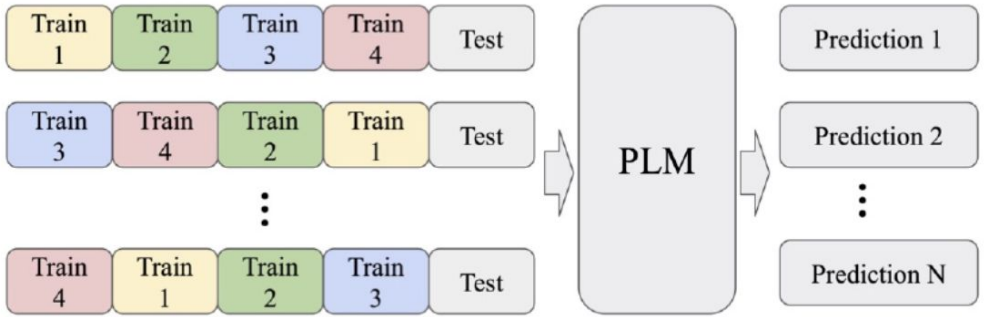
Prompt	Label Names
Review: This movie is amazing! Answer: Positive Review: Horrific movie, don't see it. Answer:	Positive, Negative
Review: This movie is amazing! Answer: good Review: Horrific movie, don't see it. Answer:	good, bad
My review for last night's film: This movie is amazing! The critics agreed that this movie was good My review for last night's film: Horrific movie, don't see it. The critics agreed that this movie was	good, bad
Here is what our critics think for this month's films. One of our critics wrote "This movie is amazing!". Her sentiment towards the film was positive. One of our critics wrote "Horrific movie, don't see it". Her sentiment towards the film was	positive, negative
Critical reception [edit] In a contemporary review, Roger Ebert wrote "This movie is amazing!". Entertainment Weekly agreed, and the overall critical reception of the film was good. In a contemporary review, Roger Ebert wrote "Horrific movie, don't see it". Entertainment Weekly agreed, and the overall critical reception of the film was	good, bad
Review: This movie is amazing! Positive Review? Yes Review: Horrific movie, don't see it. Positive Review?	Yes, No



Работа с промптами: влияние последовательности промптов с примерами

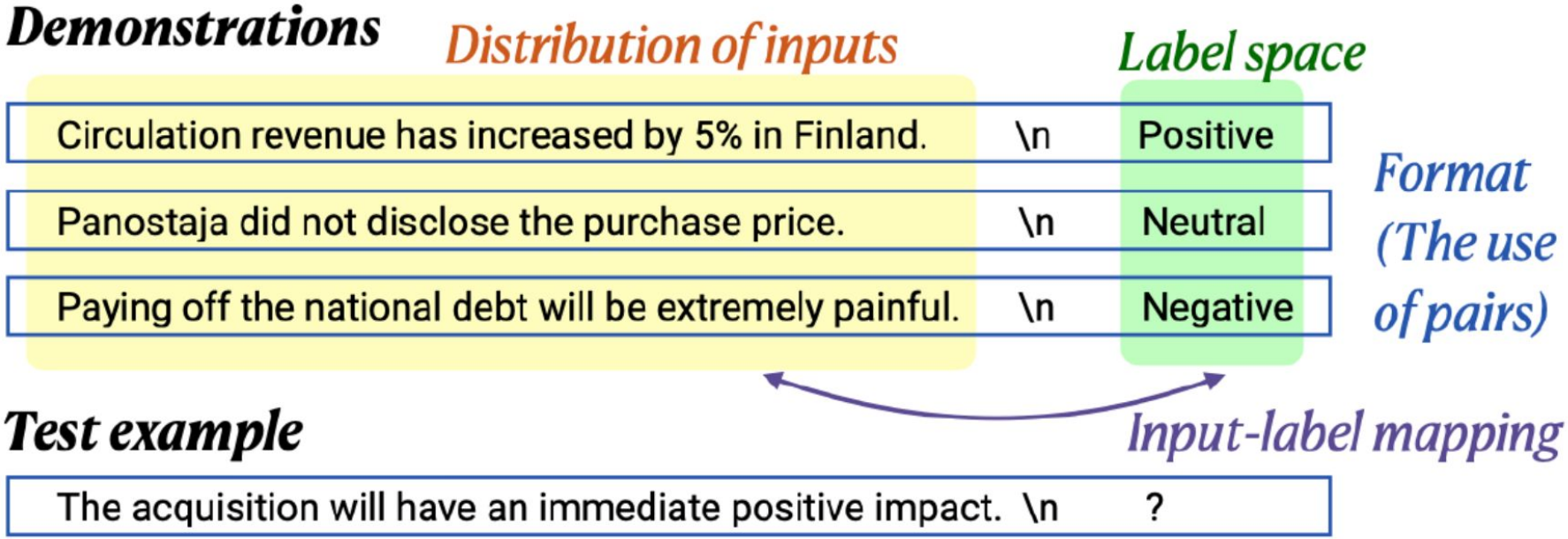
Depending on the example order in the prompt, we can get

- Near state-of-the-art accuracy
- Near random accuracy

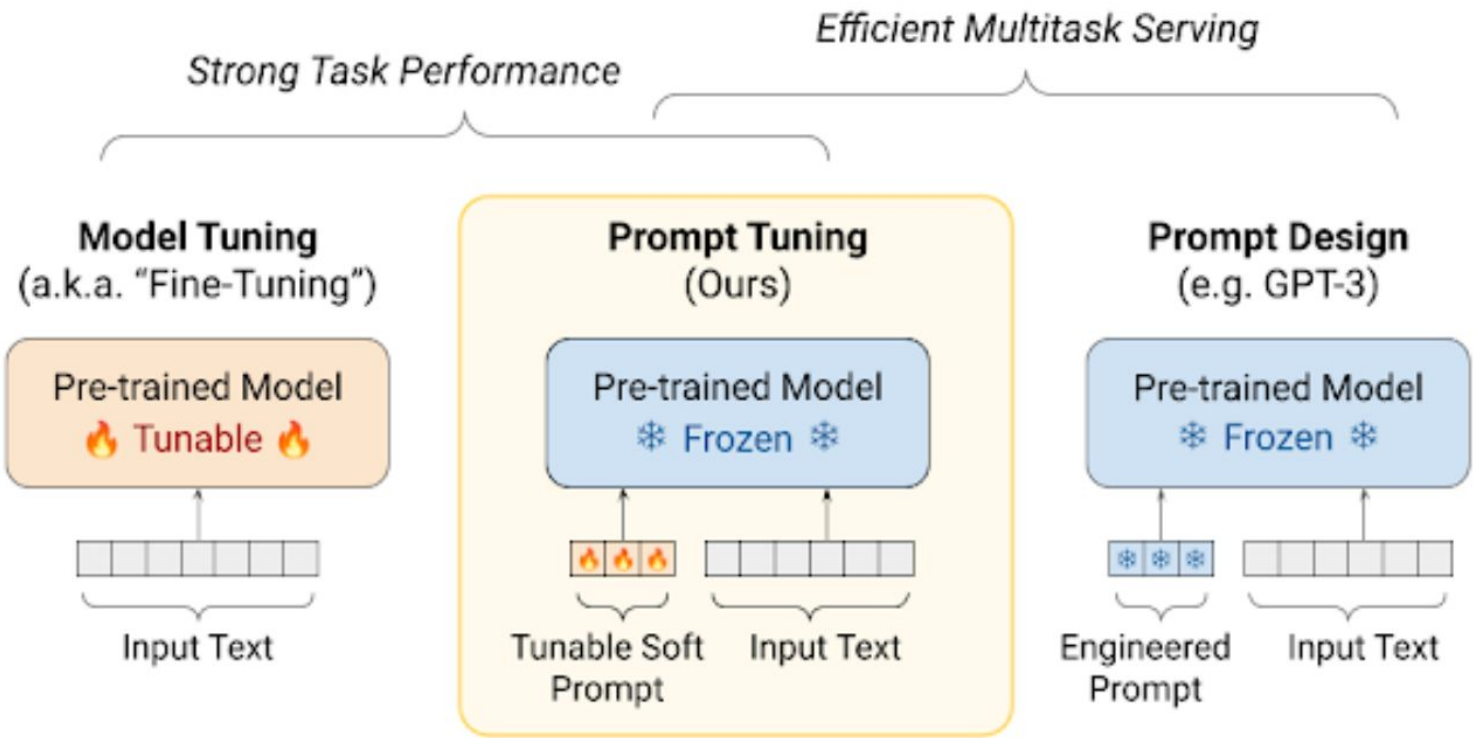


Example order is suspiciously too important

Работа с промптами: лейблы не важны? (in-context learning)



Prompt engineering и Prompt tuning



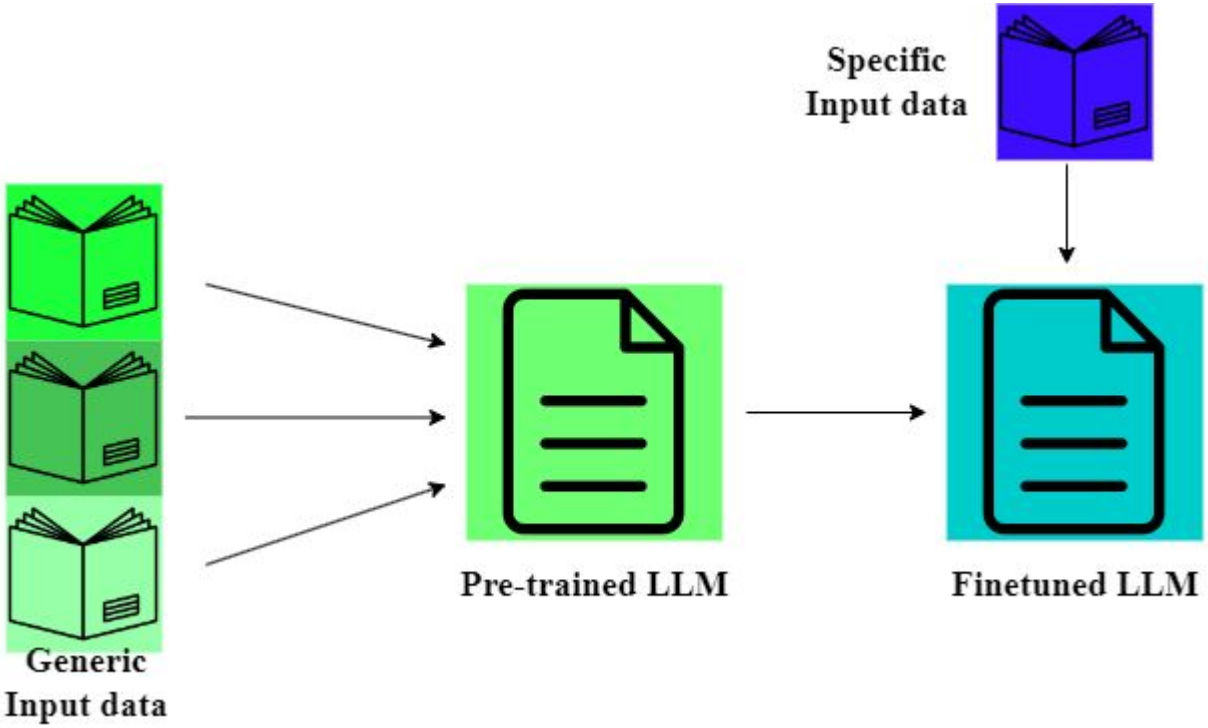
Prompt engineering и Prompt tuning

Разница между tunable soft prompt и engineered prompt:

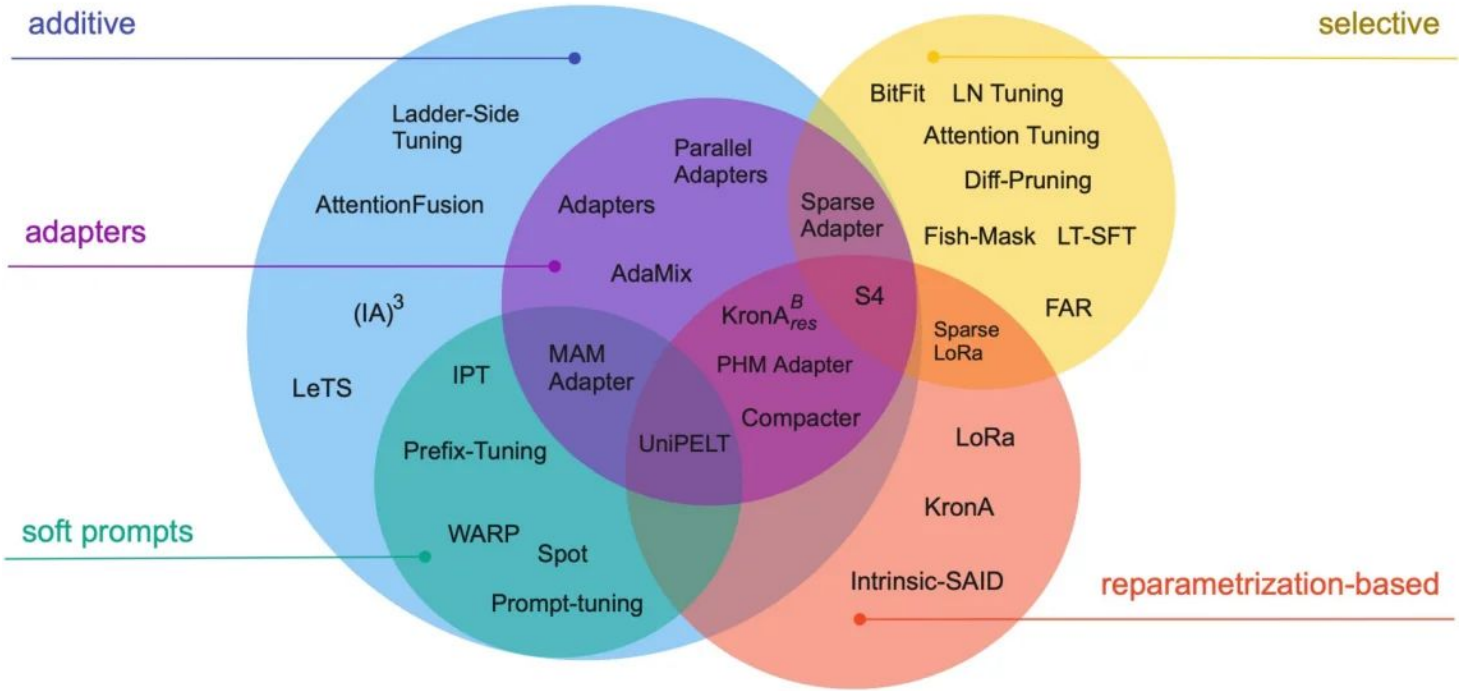
	Engineered Prompt	Tunable Soft Prompt
Тип	*Текстовый (читаемый человеком)*	*Нетребуемый к прочтению, векторный*
Создаётся	Вручную, с помощью естественного языка	Обучается, как параметры модели (но не сама модель)
Пример	`"Переведи с английского на французский:"`	Вектор размера, например, `(10, 768)`
Обучение	Не требует обучения	Требует обучения через grad descent
Интерпретируемость	Полностью интерпретируемый	Неинтерпретируемый (абстрактные эмбеддинги)
Где используется	Prompt engineering, zero/few-shot ICL	Soft Prompt Tuning, P-Tuning, Prefix Tuning и др.

PEFT (Parameter-Efficient Finetuning)

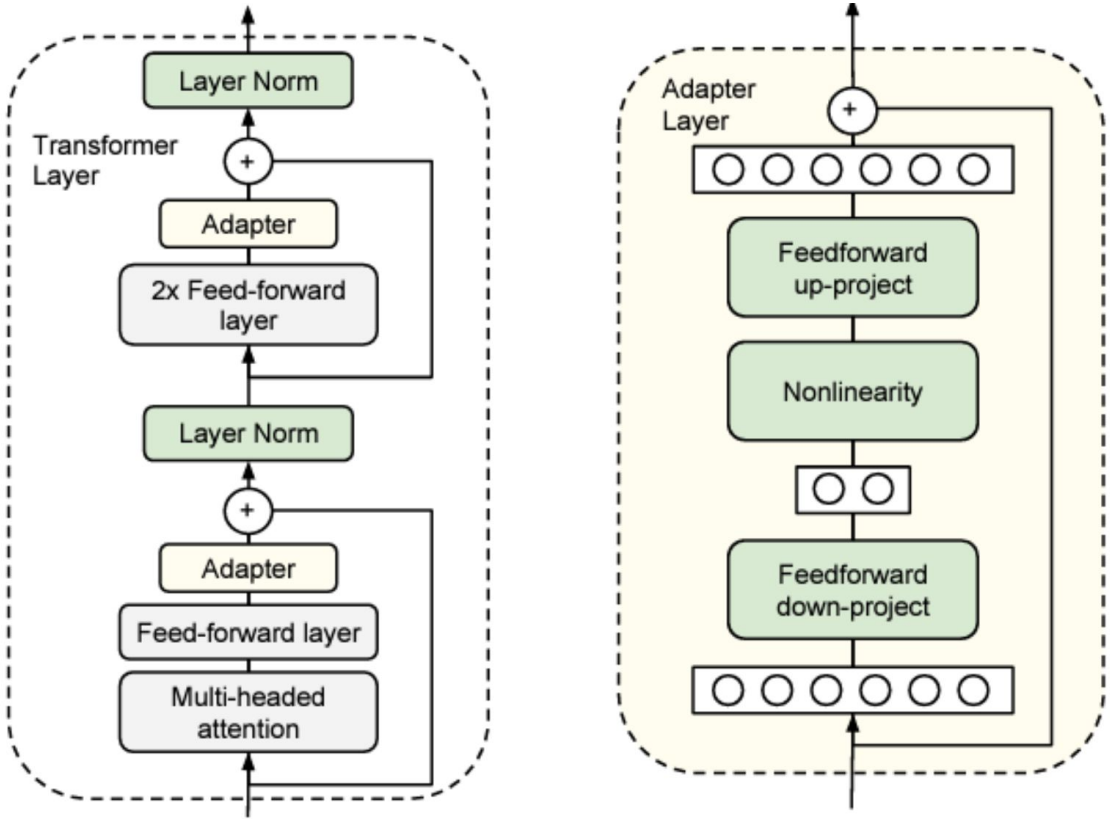
PEFT


Specific
Input data
Generic
Input data
Pre-trained LLM
Finetuned LLM

Виды PEFT


additive
selective
adapters
soft prompts
reparametrization-based
Ladder-Side Tuning
AttentionFusion
(IA)3
LeTS
IPT
Prefix-Tuning
WARP
Spot
Prompt-tuning
Parallel Adapters
Adapters
AdaMix
MAM Adapter
UniPELT
Sparse Adapter
KronA B res
PHM Adapter
Compacter
S4
BitFit
LN Tuning
Attention Tuning
Diff-Pruning
Fish-Mask
LT-SFT
FAR
Sparse LoRa
LoRa
KronA
Intrinsic-SAID

Adapters


Layer Norm
Transformer Layer
+
Adapter
2x Feed-forward layer
Layer Norm
+
Adapter
Feed-forward layer
Multi-headed attention
Adapter Layer
+
Feedforward up-project
Nonlinearity
Feedforward down-project

Low-rank adapter (LORA)

Пусть W_0 — матрица весов предобученной модели размером $n \times m$. В LoRA предлагается представить изменённые веса W в виде

$$W = W_0 + \Delta W,$$

где ΔW — это изменение весов, которое можно аппроксимировать как произведение двух низкоранговых матриц:

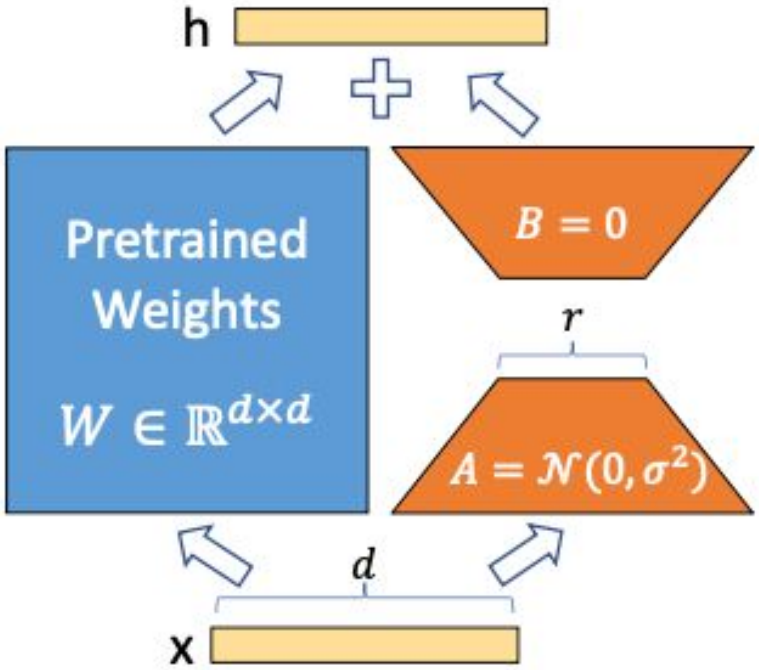
$$\Delta W = A \times B,$$

Изначально предполагаем, что в A случайные числа около 0, а B = 0

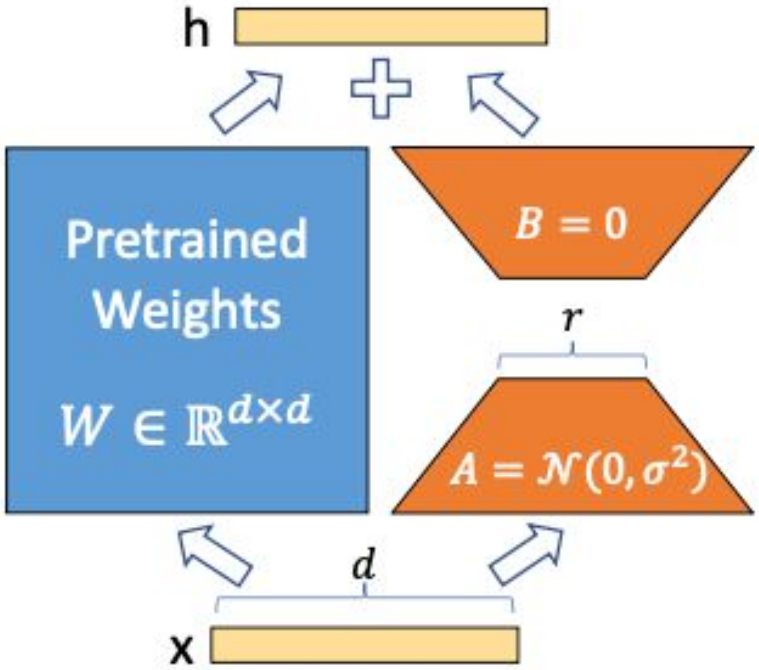
где A — матрица размера $n \times r$, а B — матрица размера $r \times m$, и r — ранг аппроксимации (обычно небольшой).

Давайте посчитаем параметры. Пусть $n = 100$, $m = 100$ тогда параметров 10^4. Теперь рассмотрим вариант с $r = 5$, тогда параметров будет $500 + 500$, всего 1000. Мы уменьшили количество параметров, которые мы вынуждены пересчитывать в 10 раз!

Low-rank adapter (LORA)



Low-rank adapter (LORA)



IA3

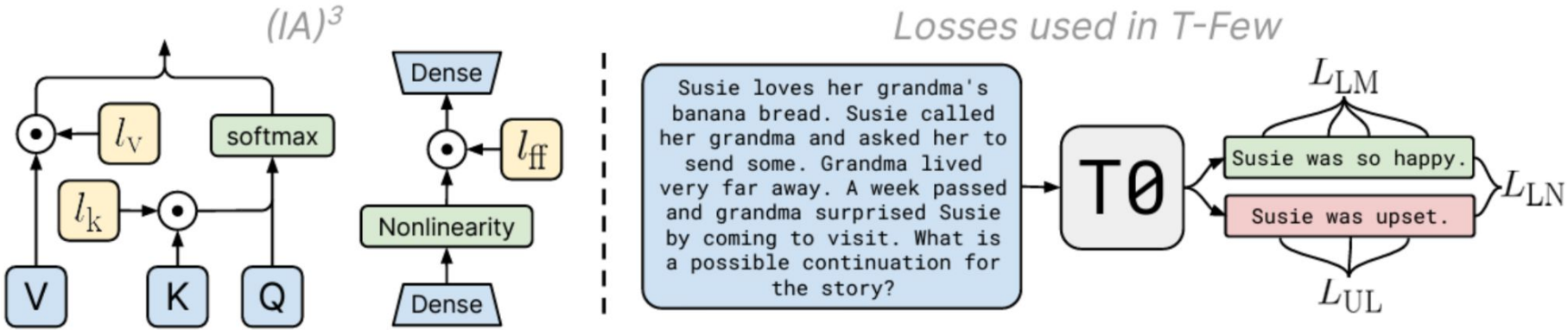
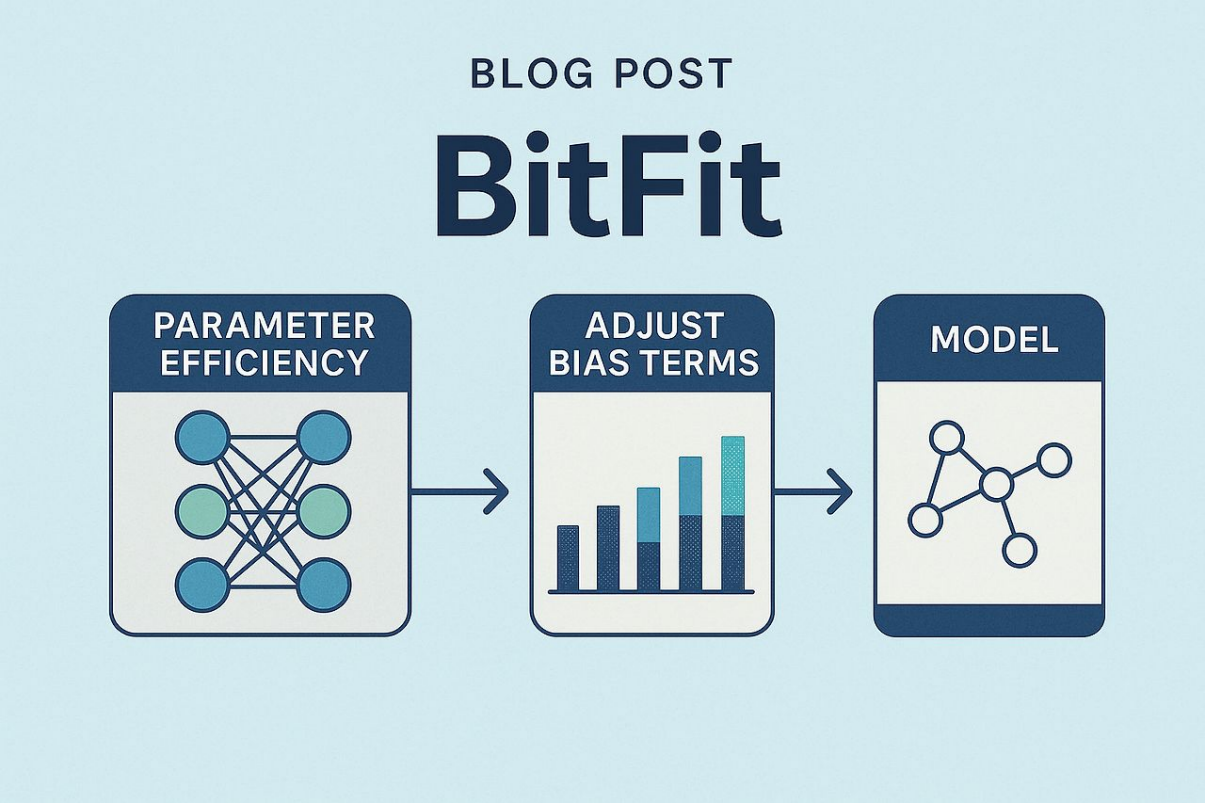


Figure 1: Diagram of (IA)^3 and the loss terms used in the T-Few recipe. *Left:* (IA)^3 introduces the learned vectors l_k, l_v, and l_{ff} which respectively rescale (via element-wise multiplication, visualized as $\odot$) the keys and values in attention mechanisms and the inner activations in position-wise feed-forward networks. *Right:* In addition to a standard cross-entropy loss L_{LM}, we introduce an unlikelihood loss L_{UL} that lowers the probability of incorrect outputs and a length-normalized loss L_{LN} that applies a standard softmax cross-entropy loss to length-normalized log-probabilities of all output choices.

BitFit


BLOG POST
BitFit
PARAMETER EFFICIENCY
ADJUST BIAS TERMS
MODEL

Сравнение методов по параметрам и качеству

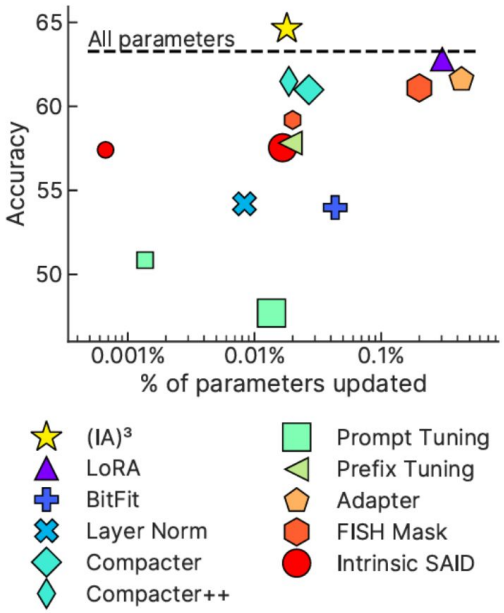


Figure 2: Accuracy of PEFT methods with L_{UL} and L_{LN} when applied to T0-3B. Methods that with variable parameter budgets are represented with larger and smaller markers for more or less parameters.

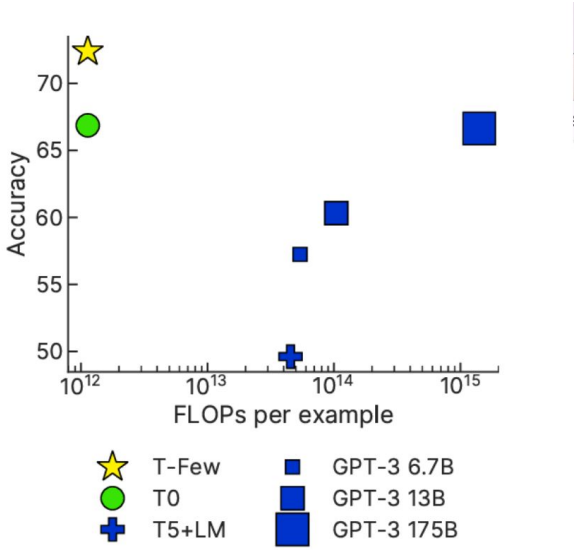


Figure 3: Accuracy of different few-shot learning methods. `T-Few` uses `(IA)`^{3} for PEFT methods of T0, T0 uses zero-shot learning, and T5+LM and the GPT-3 variants use few-shot ICL. The x-axis corresponds to inference costs; details are provided in section 4.2.

Когда что использовать?

- 10-100 примеров - prompt engineering, IA3
- 100-1000 примеров - prompt tuning, IA3
- Больше примеров - LoRA

Практика

Практика

https://colab.research.google.com/drive/11AKfq7uRYnthcSiTfkGE5fkHCVF3DjVa?usp=sharing